



MUUNTAJA

Easy User Manual

What this tool is for

Muuntaja is an ArcGIS Pro add-in that helps you import spatial files into your project and export ArcGIS layers to common delivery formats. It is designed for everyday GIS/CAD handover work: choose the data, choose the target, run, and check the messages.

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Quick Start

Import data	Export data
1. Open ArcGIS Pro and your project. 2. Click Add-In > Muuntaja. 3. Choose Tuonti. 4. Select one or more files, or a folder. 5. Pick an output GDB or folder. 6. Run and review the messages.	1. Open ArcGIS Pro and select layers or feature classes. 2. Click Add-In > Muuntaja. 3. Choose Vienti. 4. Select output folder and format. 5. Set optional CAD text/attribute options. 6. Run and find the new file in the output folder.
Best first run	
Try the tool with one small layer or one small CAD/DFSU file first. Once the output looks right, run the same settings on larger batches.	

What Muuntaja Can Do

Area	Formats / behavior
Import inputs shown in the tool	.gpkg, .geojson, .json, .kml, .kmz, .gpx, .dwg, .dxf, .dfs, or a folder containing supported files
Export outputs	GPKG, DWG, DXF, GeoJSON, Shapefile, KML, KMZ
Batch handling	Multiple import files can be processed in one run. Multiple export layers can be selected together.
Map behavior	Imported and exported outputs are added back to the active ArcGIS Pro map when possible.
Overwrite behavior	The tool creates a unique name when an output already exists, so existing files are not intentionally overwritten.
Build note	
In the current source, CAD, GeoPackage, DFSU, and generic ArcGIS-readable feature copying contain the main implementation. GPX/KML/GeoJSON import handlers are present as placeholders in this repository version, so verify those imports in your installed build before using them for production work.	

Before You Start

1. Use ArcGIS Pro 3.5 or newer, matching the add-in metadata in this project.
2. Open or create an ArcGIS Pro project before running Muuntaja.
3. Make sure the output geodatabase or output folder is writable.
4. For DFSU import, the ArcGIS Pro Python environment needs the mikeio package. The tool tries to install it automatically; if installation succeeds but import still fails, restart ArcGIS Pro and run again.
5. For CAD work, know the source coordinate system if automatic detection is uncertain. Finnish CRS choices are available in the dropdown.

Open The Tool

6. Open ArcGIS Pro.
7. Open the project and map where you want the result to appear.
8. Go to the Add-In tab.
9. Click the large Muuntaja button. The Muuntaja geoprocessing tool opens in the Geoprocessing pane.

If the button opens an error

The add-in expects Toolboxes\Muuntaja.pyt to be packaged beside the add-in DLL. Rebuild or reinstall the add-in if the toolbox is missing.

Import Workflow

Use This When

10. You received CAD, DFSU, GeoPackage, or another supported spatial file and want it in your ArcGIS project.
11. You want the output saved to the project default geodatabase or to a chosen folder.
12. You want the result added to the current map automatically.

Steps

13. Set Mitä haluat tehdä? to Tuonti.
14. In Syöte - valitse tiedosto tai kansio, choose one or more files. You can also choose a folder; the folder scan is not recursive.
15. Set Tuonti: Tallennuspaikka to a file geodatabase or folder. If you leave it empty, the project default geodatabase is used when available.
16. For CAD files, review the CAD coordinate and cleanup options.
17. For DFSU files, decide whether to filter rows before import.
18. Click Run. Watch the geoprocessing messages for progress, warnings, and output names.

CAD Import Options

Option	Meaning	Plain-language guidance
Clean CAD layers	Removes common unwanted CAD layers such as 0 and Defpoints and keeps a smaller attribute set.	Turn on when you want a cleaner GIS result. Leave off when you need all CAD detail.
[CAD] Lähtökoordinaatisto	Source coordinate system. Automaattinen tries to detect common Finnish coordinate systems.	Use Automaattinen for normal Finnish CAD data. Choose an EPSG option if you know it.
[Valinnainen] Kohde-CRS	Optional target coordinate system for projection.	Leave empty to keep the original CRS. Set it when the project requires a specific CRS.

DFSU Import With Filter

DFSU import reads model elements, detects whether they are points, lines, or polygons, creates fields for DFSU items, and can keep only matching elements.

Filter field	What to enter
[DFSU] Suodata rivejä?	Turn on to filter elements before they are written.
[DFSU] Suodatettava sarake	Choose the DFSU item/column to test.
[DFSU] Operaattori	Use =, !=, >, >=, <, <=, contains, starts with, or ends with.
[DFSU] Arvo tai teksti	Enter the numeric value or text to compare against.
Example DFSU filter	
To import only flooded elements above a threshold, enable filtering, choose the water-depth item, use operator >, and enter the threshold value, for example 0.1.	

Export Workflow

Use This When

19. You have one or more ArcGIS feature layers or feature classes and need delivery files.
20. You need a CAD handover with layer/color/text support.
21. You want a GeoPackage, Shapefile, GeoJSON, KML, KMZ, DWG, or DXF output.

Steps

22. Set Mitä haluat tehdä? to Vienti.
23. In Syöte, choose one or more feature layers or feature classes. Do not mix files for import with layers for export in the same run.
24. Choose Vienti: Vientikansio. The folder must already exist.
25. Choose Vienti: Vientiformaatti.
26. For DWG/DXF, optionally choose a text field and attribute table fields.
27. Click Run. The tool writes a uniquely named output and adds it to the map when possible.

Multiple Layers

Export format	When several layers are selected
DWG / DXF	All selected sources are prepared and exported into one CAD file.
GPKG	Selected sources become layers inside one GeoPackage.
GeoJSON / Shapefile / KML / KMZ	The tool writes separate output files for each selected layer.

CAD Export Extras

Option	What it does	When to use it
Text/label field	Creates CAD text from the selected field.	Use when labels must be visible in AutoCAD.
Create attribute table in DWG	Builds a visual table in the CAD file using lines and text.	Use for small-to-medium tabular handovers where the table should be visible in CAD.
Fields exported to table	Chooses which fields appear in the CAD table.	Pick only the fields the recipient needs; wide tables are harder to read.
CAD behavior		
The tool prepares CAD fields, uses map symbology where possible, exports points as CAD POINT entities, and sets point display size based on symbol size. Output type is DWG_R2018 or DXF_R2018.		

Parameter Reference

Parameter	Mode	Required?	Simple explanation
Mitä haluat tehdä?	Both	Yes	Switches between import and export.
Syöte	Both	Yes	Import: files/folder. Export: feature layers/classes.
Tuonti: Tallennuspaikka	Import	Optional	Output geodatabase or folder. Defaults to project GDB when possible.
CAD cleanup	Import	Optional	Filters out common unwanted CAD layers and trims fields.
[CAD] Lähtökoordinaatisto	Import	Optional	Source CRS for CAD; can be automatic.
[Valinnainen] Kohde- CRS	Import/Export prep	Optional	Projects output to another coordinate system when set.
Vienti: Vientikansio	Export	Yes	Existing folder where files are written.
Vienti: Vientiformaatti	Export	Yes	Output type: GPKG, DWG, DXF, GeoJSON, Shapefile, KML, KMZ.
CAD label field	Export	Optional	Field used to create visible CAD text.
CAD attribute table	Export	Optional	Creates a visible table inside DWG/DXF.
DFSU filter fields	Import	Optional	Keeps only DFSU elements matching a condition.

Good Habits

28. Use short, clear layer names before export; they become part of output names and CAD/GeoPackage layer names.
29. Keep field names simple for CAD recipients. Avoid very long names and special characters where possible.
30. Check the geoprocessing messages after every run. They tell you the actual saved name and any skipped empty layers.
31. For CAD import, confirm the coordinate system visually after import. Automatic detection is helpful, not magic.
32. For Shapefile export, remember normal Shapefile limitations: short field names, multiple sidcar files, and one geometry type per file.

Troubleshooting

Problem	Likely cause	What to do
The Muuntaja button cannot find the toolbox	Muuntaja.pyt was not packaged beside the DLL.	Rebuild/reinstall the add-in and confirm Toolboxes\Muuntaja.pyt is included as content.
Input is rejected	Import files and export layers are mixed, or the extension/type is unsupported.	Run import and export separately. Choose only supported files or only feature layers/classes.
Export folder error	The selected folder does not exist.	Create the folder first, then choose it in the tool.
DFSU import fails because mikeio is missing	The ArcGIS Pro Python environment cannot import mikeio.	Let the tool install it. If it says installation finished but import still fails, restart ArcGIS Pro.
CAD appears in the wrong place	Wrong or unknown source CRS.	Run again with the correct Lähtökoordinaatisto or set a target CRS.
CAD output has too much clutter	CAD cleanup was off or the source CAD has many layers.	Enable cleanup on import, or filter/clean the ArcGIS layer before export.
No rows after DFSU filter	The filter is too strict or uses the wrong item/column.	Check item names and try a wider threshold. The messages include filter diagnostics.

Glossary

Term	Meaning
GDB	File geodatabase, the normal ArcGIS workspace for storing feature classes.
Feature class	A GIS dataset containing points, lines, polygons, or other geometry.
CRS	Coordinate reference system. It defines where coordinates appear on the earth.
DFSU	DHI model file format often used for hydrologic or flood model results.

DWG / DXF	CAD formats commonly used with AutoCAD and design workflows.
GeoPackage	A single-file GIS package that can hold multiple layers.

One-page Cheat Sheet

Need	Choose	Remember
Bring a CAD file into ArcGIS	Tuonti + .dwg/.dxf	Set source CRS or use Automaattinen. Enable cleanup for cleaner layers.
Bring DFSU model data into ArcGIS	Tuonti + .dfsu	Install/restart if mikeio is needed. Use filter for large files.
Send layers to CAD	Vienti + DWG/DXF	Optional label field creates visible CAD text.
Send many layers in one GIS file	Vienti + GPKG	Each source becomes a layer inside one package.
Send a simple web/GIS exchange file	Vienti + GeoJSON/KML/KMZ	Multiple selected layers become separate files.